Pulkit **Verma**

verma.pulkit@asu.edu | pulkitverma.net | pulkitverma25 | pulkitverma25

Research Interests

AI Safety, AI Planning, Analysis of Abstractions, and Action-Model Learning.

Education

Ph.D in Computer Science, Arizona State University *Tempe, AZ*
Advisor: Prof. Siddharth Srivastava *Expected Fall 2023*

Master of Technology in Computer Science & Engineering, IIT Guwahati *Guwahati, India*
Advisor: Prof. Pradip K. Das | *Thesis*: Resource Usage Analysis for Speech Recognition Techniques. *May 2015*

Bachelor of Engineering in Information Technology, SVITS, Indore *Indore, India*
Advisor: Prof. Anand Rajawat | *Capstone*: Smart Analyzer - Predict and Cache Webpages based on Usage. *June 2011*

Publications

Pulkit Verma, and Siddharth Srivastava. "Learning Generalized Models by Interrogating Black-Box Autonomous Agents". In *AAAI 2020 Workshop on Generalization in Planning*, 2020. [Current Version]

Pulkit Verma, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016.

Pulkit Verma, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015.

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. *A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots. 2nd International Conference on Advances In Robotics*, 2015.

Pulkit Verma, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014.

Professional Experience

Arizona State University *Tempe, AZ*
Graduate Research Associate *Aug. 2018 - (present)*

- Investigating the minimal set of interfaces in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive the human interpretable model of an autonomous agent.
- Reduced the STRIPS-model comparison to a planning problem which enabled identifying if two models are not same.
- Modified FF-planner to take PDDL models, initial state, and plan as input; and output the final state.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

NetApp Inc. *Bengaluru, India*
Storage Efficiency Developer *Jul. 2015 - Jul. 2018*

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if files and volumes are cloned or modified when cloud backup is offline.

Tata Consultancy Services Ltd. *Pune, India*
iOS Application Developer *Mar. 2012 - Aug. 2013*

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

Teaching Experience

CSE 571 Artificial Intelligence [Graduate-level Course]

Arizona State University — *Tempe, AZ* — *Fall 2019*

- Delivered tutorials on Constraint Satisfaction Problems, PDDL, First Order Logic, and Dynamic Bayesian Networks.
- Co-created the software testbed used to support the course projects. Also created and graded homework assignments and exams.
- Mentored project groups working on class projects.

CS 566 Speech Processing [Graduate-level Course]

Indian Institute of Technology Guwahati — *Guwahati, India* — *Fall 2014*

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on class projects and homeworks.
- Graded homework assignments and exams.

Undergraduate Lab. Courses

Indian Institute of Technology Guwahati — *Guwahati, India* — *2013-2015*

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Graded homework assignments and exams.

Academic Projects

Analysis of Gated Updates in Value Iteration Networks

CSE 574 | Planning and Learning Methods in AI | ASU

- Replaced non gated RNN update in Value Iteration Networks with gated updates; long short term memory (LSTM), gated recurrent units (GRUs), and peephole-LSTM.
- Showed that peephole-LSTM performs best for time-bound scenarios, whereas GRUs are recommended for non-time-bound scenarios.

Automatic Image Captioning using Topic Modeling

CSE 569 | Fundamentals of Statistical Learning | ASU

- Generated image caption topics using correspondence Latent Dirichlet Allocation with SIFT image features.
- Used Gibbs sampling to calculate posteriors instead of variational inference.
- Tested on Corel-5K , ESP game, and MIR flickr datasets.
- Showed that the accuracy increases as we increase number of topics till a value, and then starts increasing again.

Artificial Immune System Based Obstacle Avoidance Robot

CS 666 | Mobile Robotics | IIT Guwahati

- Designed a robot which learns to avoid the “dynamic obstacles” using “Clonal Selection” (Immune System concept similar to Reinforcement Learning).
- Experimented with 3 scenarios; 2 robots following fixed paths, 1 robot with one periodically moving obstacle, 2 robots with unconstrained motion within a region.
- Showed that reward shaping is necessary for robots to learn to avoid obstacles in constrained environments.

Robot Cooperation through Stigmergic Communication

CS 666 | Mobile Robotics | IIT Guwahati

- Modeled the robots to perform sequential tasks by automatic-synchronization using stigmergic communication.
- Used the “Typhon” mobile agent framework to create a two level decentralized “ring” network.
- Sequentially increased the homogeneous cooperating robots in the network till task was completed autonomously.
- Demonstrated the limitations of scalability of “Typhon” which were then addressed in its next version “Tartarus”.

Resource Usage Analysis for Speech Recognition Techniques

Masters Thesis | IIT Guwahati

- Analyzed the impact of i-vectors (a projection of speech feature into total variability space) on devices having low resources like memory, storage space and power for real time speech recognition.
- Showed that i-vectors are better as compared to GMM-UBM and Joint Factor Analysis based approaches for low resource devices.

Predicting Meal Times for Automated Insulin Delivery

CS 565 | Mobile Computing | ASU

- Correlated the noisy real data sampled at different frequencies from continuous glucose monitors (CGMs) and inulin bolus.
- Experimented with SARIMA and RNN for training the time series models for glucose level prediction. Then used the predicted glucose levels and insulin bolus data to infer meal intake times.
- Showed that SARIMA predicts glucose levels with better accuracy than RNN in lesser time.

Smart Analyzer

B.E. Capstone Project | SVITS, Indore

- Designed a system that analyzed the patterns of access of websites through a Squid proxy server, and predicted the webpages to be cached.
- Used Apriori algorithm to find correlation between the webpages accessed over different time periods and predict the webpages to be cached depending on the current webpage requested.

Honors & Awards

Oct 2018	**Fulton Schools Experiential Learning Grant**, Arizona State University
Aug 2013 - May 2015	**Post Graduation Fellowship**, All India Council for Technical Education, Government of India

Other Affiliations

Nov 2020 - (present)	**Affiliated Graduate Student**, Center for Human-Compatible Artificial Intelligence, UCB
Aug 2018 - (present)	**Graduate Student**, Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU
Jul 2016 - Dec 2016	**Online Contributor**, Stanford Scholar Initiative
Jan 2014 - May 2015	**Graduate Student**, Robotics Lab, IIT Guwahati
Sep 2014	**Volunteer**, National Workshop on GPU Programming and Applications, IIT Guwahati

References

Prof. Siddharth Srivastava `siddharths@asu.edu`	Assistant Professor, School of Computing, Informatics, and Decision Systems Engineering, Arizona State University, USA
Prof. Pradip K. Das `pkdas@iitg.ac.in`	Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India
Prof. Shivashankar B. Nair `sbnair@iitg.ac.in`	Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India